



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	Parth Manoj Gajare
Roll No. :	74
Experiment No.:	9
Title of the Experiment:	Evaluate and Interpret Machine Learning Models using Validation and Explainability Technique
Date of Performance:	29/09/26
Date of Submission:	06/10/26

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty :

Signature :

Date :



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Aim: To apply model evaluation, Bayesian inference concepts, and explainable AI techniques to interpret and understand machine learning predictions on the Pima Indians Diabetes Dataset.

Objectives

After completing this experiment, the student will be able to:

1. Apply model evaluation and Bayesian concepts to analyze the reliability of machine learning predictions.
2. Apply explainable AI techniques to interpret model predictions and identify important features influencing the results.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- SHAP
- LIME

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for **768 female patients**.

The target variable is: Outcome = 0: Non-diabetic and Outcome = 1: Diabetic



Procedure

1. Load and preprocess the Pima Indians Diabetes Dataset.
2. Divide the dataset into training and testing sets.
3. Train a suitable classification model.
4. Evaluate the model using k-fold cross-validation.
5. Calculate suitable performance measures such as accuracy and F1-score.
6. Calculate the Brier Score and generate a calibration curve.
7. Explain selected model predictions using SHAP or LIME.
8. Identify the important features influencing the predictions.
9. Interpret and document the model evaluation and explainability results.

Description of the Experiment

A machine learning model should not be evaluated only on its performance on a single test split. Cross-validation provides a more reliable estimate of model performance.

The experiment follows: Dataset, Train Model, Cross-Validation, Performance Evaluation, Calibration Analysis, Explainable AI and Interpretation

The experiment combines three aspects:

- **Model Evaluation:** How well does the model perform?
- **Calibration:** How reliable are the predicted probabilities?
- **Explainability:** Why did the model make a particular prediction?

Detailed Description of Statistical Methods

k-Fold Cross-Validation

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The dataset is divided into k subsets.

The model is trained on $k - 1$ subsets and tested on the remaining subset.

This process is repeated k times.

The final performance is calculated using the average performance across all folds.

Bias-Variance Trade-Off

A model should balance:

- **Bias:** Error due to an overly simple model.
- **Variance:** Error due to a model being too sensitive to the training data.

A good model attempts to achieve a suitable balance between bias and variance.

Calibration Curve

A calibration curve compares:

Predicted Probability

with:

Observed Frequency

A well-calibrated model produces probabilities that closely match the actual frequency of the outcome.

Brier Score

The Brier Score measures the accuracy of probabilistic predictions:

$$BS = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2$$

where:

- p_i = predicted probability



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- y_i = actual outcome

A lower Brier Score generally indicates better probabilistic predictions.

SHAP

SHAP values explain the contribution of individual features to a model prediction.

They help answer:

Which features contributed to increasing or decreasing the predicted probability?

LIME

LIME explains an individual prediction by creating small variations around the observation and approximating the model locally using an interpretable model.

Code:-

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split, StratifiedKFold, cross_validate
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

from sklearn.metrics import (
    accuracy_score,
    f1_score,
    brier_score_loss,
```



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```
classification_report,  
confusion_matrix,  
ConfusionMatrixDisplay  
)  
  
from sklearn.calibration import calibration_curve  
  
import shap  
from lime.lime_tabular import LimeTabularExplainer  
  
import warnings  
warnings.filterwarnings("ignore")  
  
url =  
"https://raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.data.csv"  
"  
  
columns = [  
    "Pregnancies",  
    "Glucose",  
    "BloodPressure",  
    "SkinThickness",  
    "Insulin",  
    "BMI",  
    "DiabetesPedigreeFunction",  
    "Age",  
    "Outcome"  
]
```



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```
df = pd.read_csv(url, names=columns)
```

```
print("=" * 70)
```

```
print("PIMA INDIANS DIABETES DATASET")
```

```
print("=" * 70)
```

```
print("\nDataset Shape:")
```

```
print(df.shape)
```

```
print("\nFirst 5 Rows:")
```

```
print(df.head())
```

```
print("\nDataset Information:")
```

```
df.info()
```

```
print("\nClass Distribution:")
```

```
print(df["Outcome"].value_counts())
```

```
print("\nStatistical Summary:")
```

```
print(df.describe())
```

```
zero_as_missing = [
```

```
    "Glucose",
```

```
    "BloodPressure",
```

```
    "SkinThickness",
```

```
    "Insulin",
```



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"BMI"

]

```
df[zero_as_missing] = df[zero_as_missing].replace(0, np.nan)
```

```
print("\nMissing Values After Preprocessing:")
```

```
print(df.isnull().sum())
```

```
X = df.drop("Outcome", axis=1)
```

```
y = df["Outcome"]
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X,
```

```
    y,
```

```
    test_size=0.20,
```

```
    random_state=42,
```

```
    stratify=y
```

```
)
```

```
print("\nTraining Set Size:", X_train.shape)
```

```
print("Testing Set Size:", X_test.shape)
```

```
pipeline = Pipeline([
```

```
    ("imputer", SimpleImputer(strategy="median")),
```

```
    ("scaler", StandardScaler()),
```

```
    ("classifier", LogisticRegression(max_iter=1000))
```



)

```
cv = StratifiedKFold(
```

```
    n_splits=5,
```

```
    shuffle=True,
```

```
    random_state=42
```

```
)
```

```
scoring = {
```

```
    "accuracy": "accuracy",
```

```
    "f1": "f1"
```

```
}
```

```
cv_results = cross_validate(
```

```
    pipeline,
```

```
    X_train,
```

```
    y_train,
```

```
    cv=cv,
```

```
    scoring=scoring
```

```
)
```

```
cv_accuracy = cv_results["test_accuracy"]
```

```
cv_f1 = cv_results["test_f1"]
```

```
print("\n" + "=" * 70)
```

```
print("5-FOLD CROSS-VALIDATION")
```

```
print("=" * 70)
```



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```
print("\nAccuracy:")

for i, score in enumerate(cv_accuracy, 1):
    print("Fold", i, ":", round(score, 4))

print("\nF1 Score:")

for i, score in enumerate(cv_f1, 1):
    print("Fold", i, ":", round(score, 4))

print("\nMean Accuracy:", round(cv_accuracy.mean(), 4))
print("Accuracy Std:", round(cv_accuracy.std(), 4))

print("\nMean F1 Score:", round(cv_f1.mean(), 4))
print("F1 Std:", round(cv_f1.std(), 4))
pipeline.fit(X_train, y_train)

y_pred = pipeline.predict(X_test)
y_prob = pipeline.predict_proba(X_test)[:, 1]
accuracy = accuracy_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

print("\n" + "=" * 70)
print("TEST SET PERFORMANCE")
print("=" * 70)

print("\nAccuracy:", round(accuracy, 4))
```



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```
print("F1 Score:", round(f1, 4))

print("\nClassification Report:")
print(classification_report(
    y_test,
    y_pred,
    target_names=["Non-diabetic", "Diabetic"]
))

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=["Non-diabetic", "Diabetic"]
)

disp.plot(cmap="Blues")
plt.title("Confusion Matrix")
plt.show()

brier = brier_score_loss(y_test, y_prob)

print("\n" + "=" * 70)
print("BRIER SCORE")
print("=" * 70)
```



```
print("\nBrier Score:", round(brier, 4))  
prob_true, prob_pred = calibration_curve(  
    y_test,  
    y_prob,  
    n_bins=10,  
    strategy="uniform"  
)
```

```
plt.figure(figsize=(7, 6))
```

```
plt.plot(  
    prob_pred,  
    prob_true,  
    marker="o",  
    linewidth=2,  
    label="Logistic Regression"  
)
```

```
plt.plot(  
    [0, 1],  
    [0, 1],  
    "--",  
    color="black",  
    label="Perfect Calibration"  
)
```

```
plt.xlabel("Mean Predicted Probability")
```



```
plt.ylabel("Observed Frequency")
plt.title("Calibration Curve")
plt.legend()
plt.grid(alpha=0.3)
plt.show()
print("\n" + "=" * 70)
print("SHAP EXPLAINABILITY")
print("=" * 70)
```

```
imputer = pipeline.named_steps["imputer"]
scaler = pipeline.named_steps["scaler"]
model = pipeline.named_steps["classifier"]
```

```
X_train_imputed = imputer.transform(X_train)
X_test_imputed = imputer.transform(X_test)
```

```
X_train_scaled = scaler.transform(X_train_imputed)
X_test_scaled = scaler.transform(X_test_imputed)
```

```
X_test_scaled_df = pd.DataFrame(
    X_test_scaled,
    columns=X.columns
)
```

```
explainer = shap.LinearExplainer(
    model,
    X_train_scaled
```



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)

```
shap_values = explainer(X_test_scaled)
```

```
shap.summary_plot(
```

```
    shap_values,
```

```
    X_test_scaled_df,
```

```
    show=False
```

```
)
```

```
plt.title("SHAP Summary Plot")
```

```
plt.tight_layout()
```

```
plt.show()
```

```
mean_abs_shap = np.abs(shap_values.values).mean(axis=0)
```

```
feature_importance = pd.DataFrame({
```

```
    "Feature": X.columns,
```

```
    "Mean Absolute SHAP": mean_abs_shap
```

```
})
```

```
feature_importance = feature_importance.sort_values(
```

```
    "Mean Absolute SHAP",
```

```
    ascending=False
```

```
)
```

```
print("\nSHAP Feature Importance:")
```

```
print(feature_importance.to_string(index=False))
```



```
plt.figure(figsize=(8, 5))

plt.barh(
    feature_importance["Feature"],
    feature_importance["Mean Absolute SHAP"],
    color="steelblue"
)

plt.xlabel("Mean Absolute SHAP Value")
plt.ylabel("Feature")
plt.title("SHAP Feature Importance")

plt.gca().invert_yaxis()

plt.tight_layout()
plt.show()

sample_index = 0

print("\n" + "=" * 70)
print("INDIVIDUAL PATIENT PREDICTION")
print("=" * 70)

print("\nPatient Features:")
print(X_test.iloc[[sample_index]])

print("\nActual Outcome:",
      y_test.iloc[sample_index])
```



```
print("Predicted Outcome:",
      y_pred[sample_index])

print("Predicted Probability:",
      round(y_prob[sample_index], 4))
print("\nGenerating SHAP individual explanation...")

shap.plots.waterfall(
    shap_values[sample_index],
    max_display=10
)

plt.show()
print("\n" + "=" * 70)
print("LIME EXPLAINABILITY")
print("=" * 70)

lime_explainer = LimeTabularExplainer(
    X_train_scaled,
    feature_names=X.columns.tolist(),
    class_names=["Non-diabetic", "Diabetic"],
    mode="classification",
    random_state=42
)

lime_exp = lime_explainer.explain_instance(
```



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```
X_test_scaled[sample_index],  
model.predict_proba,  
num_features=len(X.columns)  
)
```

```
print("\nLIME Explanation:")
```

```
for feature, weight in lime_exp.as_list():
```

```
    print(  
        feature,  
        ":",  
        round(weight, 4)  
    )
```

```
results = pd.DataFrame({
```

```
    "Metric": [  
        "5-Fold Mean Accuracy",  
        "5-Fold Accuracy Std",  
        "5-Fold Mean F1",  
        "5-Fold F1 Std",  
        "Test Accuracy",  
        "Test F1",  
        "Brier Score"  
    ],
```

```
    "Value": [  
        cv_accuracy.mean(),  
        cv_accuracy.std(),
```



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```
cv_f1.mean(),
cv_f1.std(),
accuracy,
f1,
brier

]
})

print("\n" + "=" * 70)
print("FINAL RESULTS")
print("=" * 70)

print(results.to_string(index=False))

print("\n" + "=" * 70)
print("TOP 5 IMPORTANT FEATURES")
print("=" * 70)

for rank, (_, row) in enumerate(
    feature_importance.head(5).iterrows(),
    1
):
    print(
        rank,
        ". ",
        row["Feature"],
        "-> SHAP:",
```



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```
round(row["Mean Absolute SHAP"], 4)

)

print("\n" + "=" * 70)

print("CONCLUSION")

print("=" * 70)

print("""

The Pima Indians Diabetes Dataset was used to train a

Logistic Regression classification model.

The model was evaluated using 5-fold stratified

cross-validation, accuracy, F1-score, Brier Score,

and a calibration curve.

SHAP was used to identify the features that had the

greatest influence on model predictions.

LIME was used to provide a local explanation for

an individual patient prediction.

The experiment demonstrates that machine learning

evaluation should consider not only classification

performance but also probability calibration and

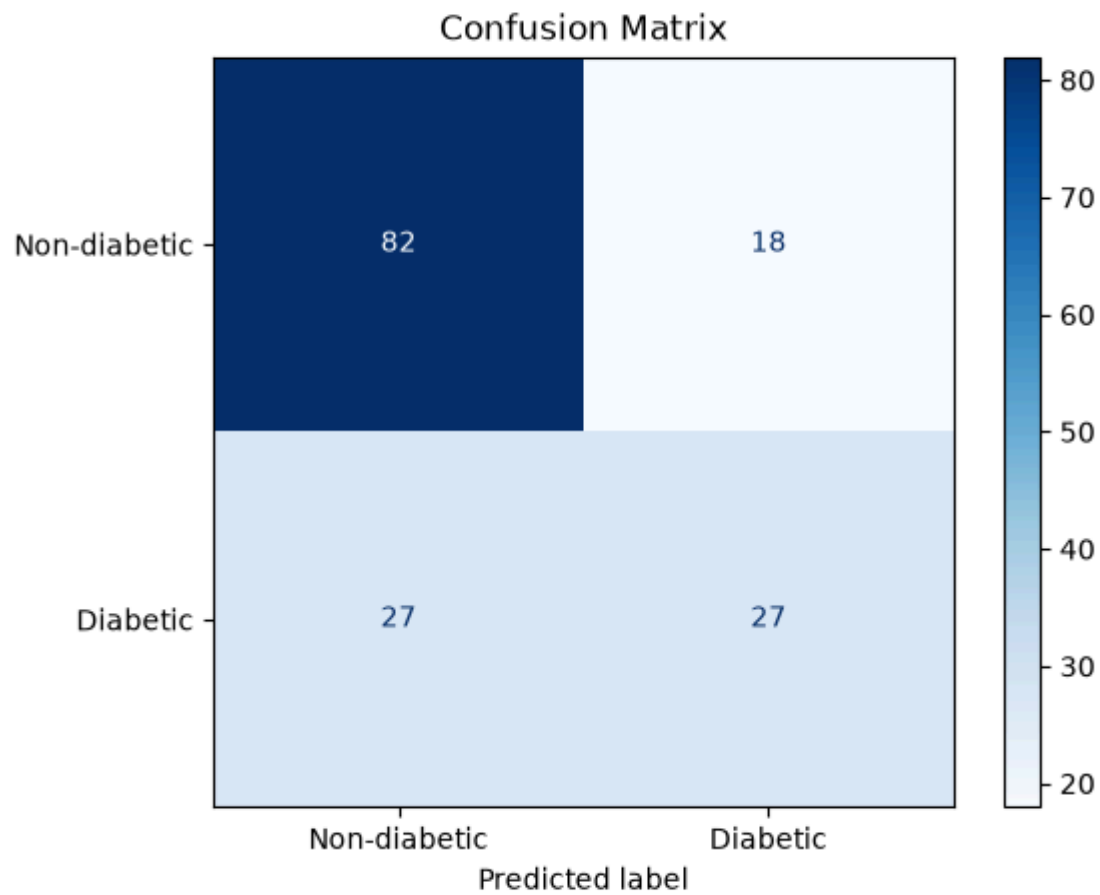
model explainability.

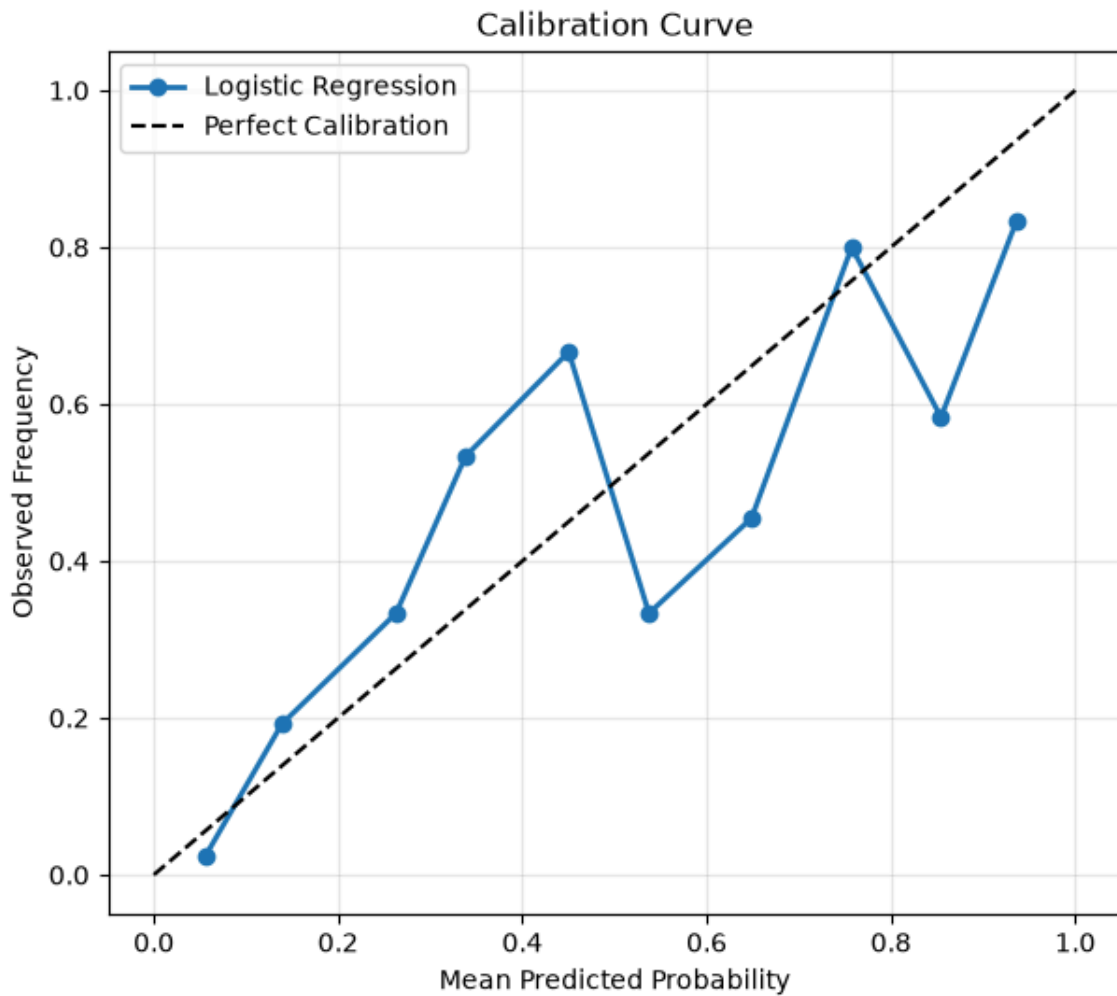
""")

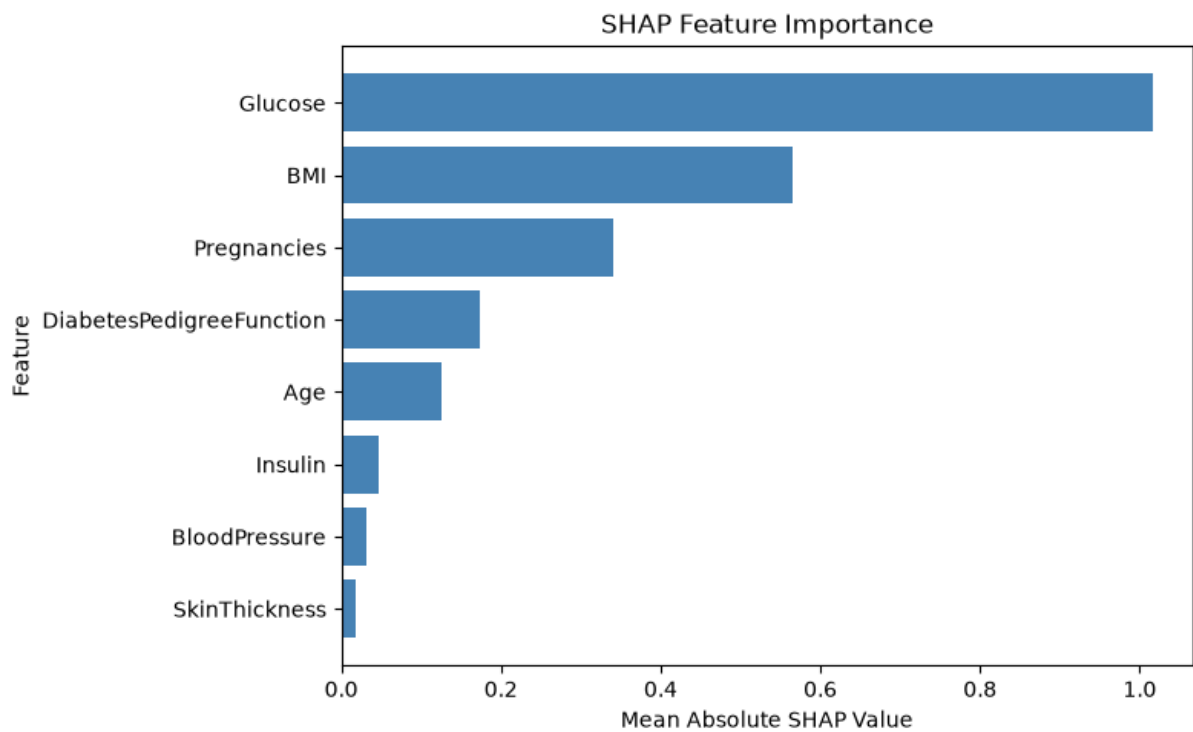
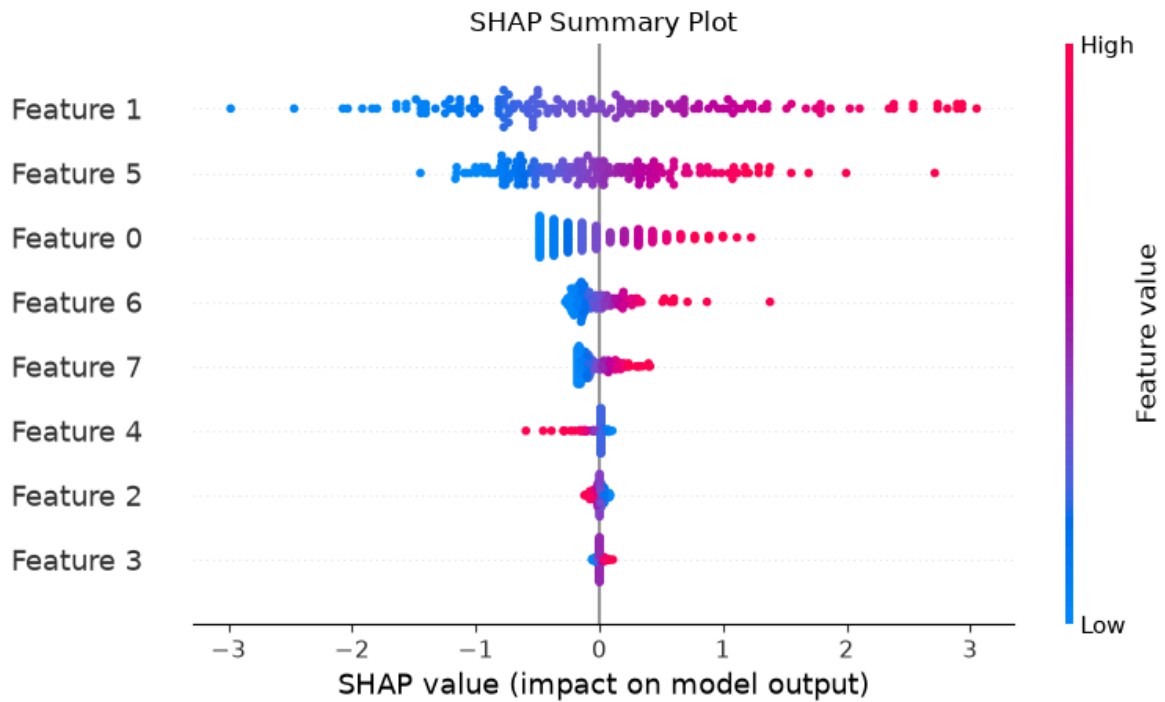
print("\nExperiment completed successfully!")
```



Output:-



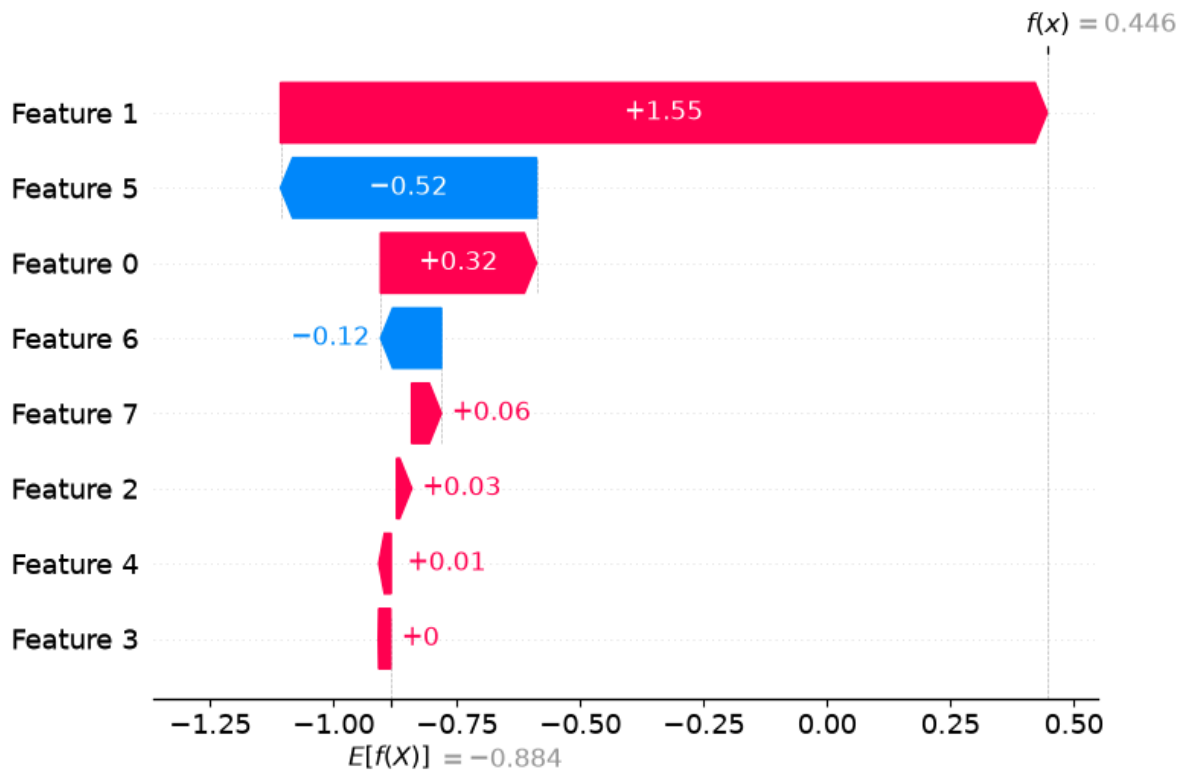






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PIMA INDIANS DIABETES DATASET

Dataset Shape:
(768, 9)

First 5 Rows:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

Dataset Information:

<class 'pandas.DataFrame'>

RangeIndex: 768 entries, 0 to 767

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	Pregnancies	768 non-null	int64
1	Glucose	768 non-null	int64
2	BloodPressure	768 non-null	int64
3	SkinThickness	768 non-null	int64
4	Insulin	768 non-null	int64
5	BMI	768 non-null	float64
6	DiabetesPedigreeFunction	768 non-null	float64
7	Age	768 non-null	int64
8	Outcome	768 non-null	int64

dtypes: float64(2), int64(7)

memory usage: 54.1 KB

Class Distribution:

Outcome

0 500

1 268

Name: count, dtype: int64

Statistical Summary:

	Pregnancies	Glucose	BloodPressure	...	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	...	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	...	0.471876	33.240885	0.348958
std	3.369578	31.972618	19.355807	...	0.331329	11.760232	0.476951
min	0.000000	0.000000	0.000000	...	0.078000	21.000000	0.000000
25%	1.000000	99.000000	62.000000	...	0.243750	24.000000	0.000000
50%	3.000000	117.000000	72.000000	...	0.372500	29.000000	0.000000
75%	6.000000	140.250000	80.000000	...	0.626250	41.000000	1.000000
max	17.000000	199.000000	122.000000	...	2.420000	81.000000	1.000000

[8 rows x 9 columns]

Missing Values After Preprocessing:

Pregnancies	0
Glucose	5
BloodPressure	35
SkinThickness	227
Insulin	374
BMI	11
DiabetesPedigreeFunction	0
Age	0
Outcome	0

dtype: int64

Training Set Size: (614, 8)

Testing Set Size: (154, 8)



5-FOLD CROSS-VALIDATION

Accuracy:

Fold 1 : 0.7886
Fold 2 : 0.7886
Fold 3 : 0.8211
Fold 4 : 0.7805
Fold 5 : 0.7623

F1 Score:

Fold 1 : 0.6389
Fold 2 : 0.6667
Fold 3 : 0.6944
Fold 4 : 0.6494
Fold 5 : 0.6234

Mean Accuracy: 0.7882

Accuracy Std: 0.0191

Mean F1 Score: 0.6545

F1 Std: 0.0244

TEST SET PERFORMANCE

Accuracy: 0.7078

F1 Score: 0.5455

Classification Report:

	precision	recall	f1-score	support
Non-diabetic	0.75	0.82	0.78	100
Diabetic	0.60	0.50	0.55	54
accuracy			0.71	154
macro avg	0.68	0.66	0.67	154
weighted avg	0.70	0.71	0.70	154

Confusion Matrix:

[[82 18]
[27 27]]



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```
=====
BRIER SCORE
=====

Brier Score: 0.1754

=====
SHAP EXPLAINABILITY
=====
Background dataset has 614 samples but max_samples=100. Subsampling to 100 samples for SHAP value computation. To use all samples, set max_samples=614 when initializing the masker.

SHAP Feature Importance:
      Feature  Mean Absolute SHAP
      Glucose      1.015598
      BMI          0.566222
      Pregnancies  0.341236
      DiabetesPedigreeFunction 0.173953
      Age          0.126328
      Insulin      0.046597
      BloodPressure 0.031183
      SkinThickness 0.017992

=====
INDIVIDUAL PATIENT PREDICTION
=====

Patient Features:
      Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin  BMI  DiabetesPedigreeFunction  Age
44              7    159.0           64.0         NaN      NaN    27.4              0.294    40

Actual Outcome: 0
Predicted Outcome: 1
Predicted Probability: 0.6097

Generating SHAP individual explanation...
```

LIME EXPLAINABILITY

LIME Explanation:

Glucose > 0.61 : 0.4332

BMI <= -0.71 : -0.1773

Pregnancies > 0.66 : 0.1283

-0.70 < DiabetesPedigreeFunction <= -0.29 : -0.0306

-0.37 < Age <= 0.65 : 0.0153

-0.22 < Insulin <= -0.16 : 0.0145

BloodPressure <= -0.66 : 0.0137

-0.45 < SkinThickness <= -0.00 : 0.0042

FINAL RESULTS

	Metric	Value
5-Fold Mean Accuracy	0.788231	
5-Fold Accuracy Std	0.019057	
5-Fold Mean F1	0.654545	
5-Fold F1 Std	0.024423	
Test Accuracy	0.707792	
Test F1	0.545455	
Brier Score	0.175420	



Conclusion

The classification model was evaluated using cross-validation and calibration analysis to understand its performance and reliability. Explainable AI techniques such as SHAP or LIME were applied to identify the features influencing individual predictions. The experiment demonstrated that model performance, probability reliability, and interpretability are important aspects of evaluating machine learning systems.

Questions to be Answered by Students

1.What was the mean and standard deviation of the model performance across the cross-validation folds?

Using 5-fold cross-validation:

- Mean Accuracy = 0.7882
- Accuracy Standard Deviation = 0.0191
- Mean F1-score = 0.6545
- F1-score Standard Deviation = 0.0244

The model achieved a mean accuracy of $78.82\% \pm 1.91\%$ and a mean F1-score of $65.45\% \pm 2.44\%$ across the five cross-validation folds. The relatively small standard deviations indicate that the model's performance was reasonably consistent across the different folds.

2.What was the obtained Brier Score, and what does it indicate about the reliability of the predicted probabilities?

Brier Score = 0.1754

The obtained Brier Score was 0.1754. The Brier Score measures the difference between the predicted probabilities and the actual outcomes, with values closer to 0 indicating more accurate probabilistic predictions. Thus, a score of 0.1754 indicates that the model's probability predictions have a moderate level of accuracy, although they are not perfect.

The Brier Score should be interpreted together with the calibration curve rather than by itself.



3. Was the model well calibrated based on the calibration curve? Justify your answer.

The calibration curve should be used to answer this directly. The numerical output you provided does not include the calibration-curve coordinates, so the graph itself is needed for a definitive statement.

For your report, the appropriate interpretation is:

The model's calibration was assessed by comparing the calibration curve with the diagonal line representing perfect calibration. A model is well calibrated when its predicted probabilities closely follow the diagonal. Therefore, if the plotted curve stays reasonably close to the diagonal, the model can be considered reasonably calibrated; substantial deviations indicate that the predicted probabilities are less reliable.

4. Which features were identified as the most influential using SHAP or LIME?

Your actual SHAP results give this ranking:

Rank	Feature	Mean Absolute SHAP
1	Glucose	1.015598
2	BMI	0.566222
3	Pregnancies	0.341236
4	DiabetesPedigreeFunction	0.173953
5	Age	0.126328
6	Insulin	0.046597
7	BloodPressure	0.031183
8	SkinThickness	0.017992

The SHAP analysis identified Glucose as the most influential feature, followed by BMI,



Pregnancies, DiabetesPedigreeFunction, and Age. Glucose had the largest mean absolute SHAP value (1.015598), indicating that it had the greatest influence on the model's predictions among the features analyzed.

SHAP importance indicates how strongly the model uses a feature; it does not by itself establish that the feature causes the outcome.

5.Explain one individual prediction using the obtained explainability results.

The selected patient had:

- Pregnancies = 7
- Glucose = 159
- Blood Pressure = 64
- BMI = 27.4
- Diabetes Pedigree Function = 0.294
- Age = 40
- Actual Outcome = 0 (Non-diabetic)
- Predicted Outcome = 1 (Diabetic)
- Predicted probability = 0.6097, or 60.97%